

Computational Fluid Dynamics - 9/1/2024

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We want to simulate the natural convection flow in a square cavity induced by the presence of a hot cylindrical object located inside the cavity. The flow is governed by the following unsteady incompressible Navier–Stokes equations with the *Boussinesq buoyancy model* on the domain $\Omega = (0, 1) \times (0, 1)$:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \nu \Delta \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p + \beta \mathbf{g}(T - T_{\text{ref}}) = 0, & \text{in } \Omega, t \in (0, t_f), \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, t \in (0, t_f), \\ \frac{\partial T}{\partial t} - k \Delta T + \mathbf{u} \cdot \nabla T = 0, & \text{in } \Omega, t \in (0, t_f), \end{cases} \quad (1)$$

where $\mathbf{u}$, p and T are velocity, pressure and temperature, respectively, ν is the kinematic viscosity, β is the thermal expansion coefficient, $\mathbf{g} = (0, -9.8)$ is the gravity acceleration, T_{ref} is a reference temperature and k the thermal diffusion coefficient.

Moreover, consider the following boundary conditions for the flow equations and for temperature evolution:

- the internal cylinder is a rough solid wall with prescribed temperature $T = 30$;
- the bottom boundary is a rough solid wall with prescribed temperature $T = 0$;
- the top and lateral boundaries are adiabatic, rough solid walls
(*Hint*: set homogeneous Neumann condition on temperature).

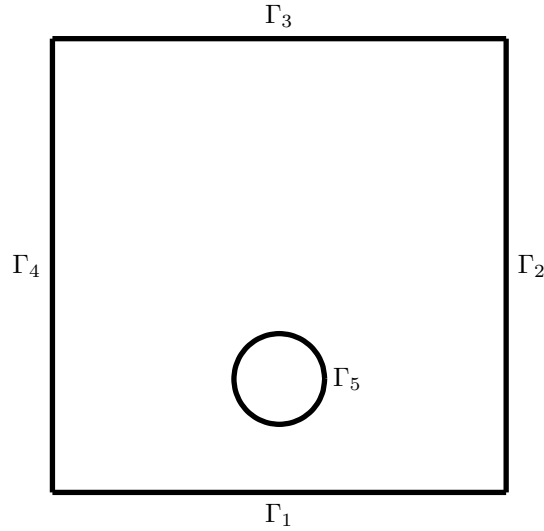


Figure 1: Computational domain

- a) Write explicitly the boundary conditions that should be prescribed on each of the domain boundaries.
- b) Introduce a suitable time-advancing scheme for the time-discretization of the proposed problem such that, at each time step, continuity and momentum equations are first solved using the incremental Chorin-Temam projection method and then the energy (temperature) equation is solved using the updated velocity field.

- c) Write the weak formulations of the problems to solve at each time iteration discussing the choice of functional spaces and the treatment of boundary conditions.
- d) Write a Firedrake notebook, based on the template `template090124.ipynb`, implementing the proposed scheme based on $\mathbb{P}_2$ finite elements for velocity and $\mathbb{P}_1$ finite elements for pressure and temperature. As initial condition, consider the flow at rest and zero temperature.
- e) Given $\nu = 0.01$, $k = 0.01$, $\alpha = 0.001$, $T_{\text{ref}} = 0$, compute the solution until the final time $t_f = 100$ with a time step $\Delta t = 1$. Then:
- save plots of the velocity and temperature fields at the final time (export them as files `SURNAME_NAME_velocity.png`, `SURNAME_NAME_temperature.png`)
 - evaluate the mean temperature value over the whole domain at the final time.